Regression Analysis on Ecuadorian Spanish Speaker's Perspective on Trill

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Intro

Research Questions

- Does hearing a trilled vs. an assibilated /r/ change a listener's perception of a speaker's social status or characteristics?
- How do listeners identify a speaker based on their trill?
- Do Lojano and non-Lojano listeners evaluate trilled speakers differently?

Data Collection

- 210 final respondents (4 listener regions: Loja, Cuenca, Coast , Quito).
- 8 Lojano speakers: 4 trilled, 4 assibilated.
- Listeners rated each speaker on femininity, niceness, confidence, class, urbanism, education, age, and predicted origin.

Data Cleaning

Pivot_longer(): expanded the 200-count dataset to nearly 2000 values.

- cols=matches("[a-z]_[a-z]", ignore.case=FALSE).
- names_to = c("Speaker",".value").
- names_sep = "_".

Mutate(): create informative columns.
Rename(): create clarifying column names.
Filter(): remove uncompleted survey attempts.

Daniela.origin	Daniela.edu	Daniela.age	Daniela.eth
2	3	3	mestiza
NA	NA	NA	
4	2	2	Negra
1	1	3	Indí_gena
4	2	3	Serrana

trill	Speaker	fem	nice	conf	class
0	F2	5	5	4	5
1	F2	2	6	4	2
0	M2	6	4	6	3
1	F1	6	5	5	3
0	M1	3	3	3	3
0	F1	3	1	1	4

Perceived Origin

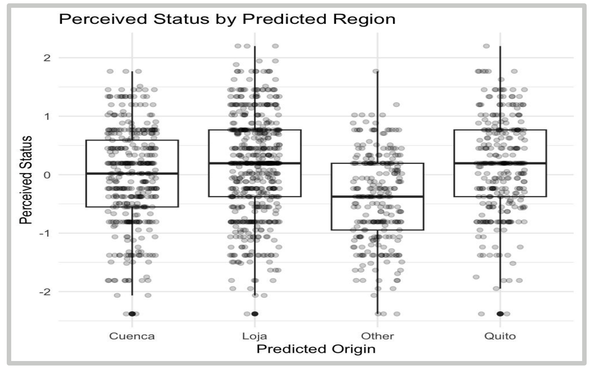
Model Preparation:
Created binary outcome variables from predictedOrigin.

Model 1: Does trill affect guessing Quito?
Trill non-significant ($p = 0.538$), status significant ($p = 0.003$), region significant ($p = 0.025$).
Finding: Trill does not push listeners towards guessing Quito.
Model 2: Does trill predict correct Loja identification?
Trill significant ($p < 0.001$), status significant ($p < 0.001$), region significant ($p < 0.001$).
Finding: Hearing a trill nearly doubled the odds of correctly identifying a speaker as being from Loja.
Model 3: Does the trill effect on Loja attribution differ by listener region?
Trill x region marginally significant ($p = 0.07$), trill:RegionCuenca significant ($p = 0.038$).
Finding: The trill's effect on correct Loja identification was strongest for Cuenca listeners.

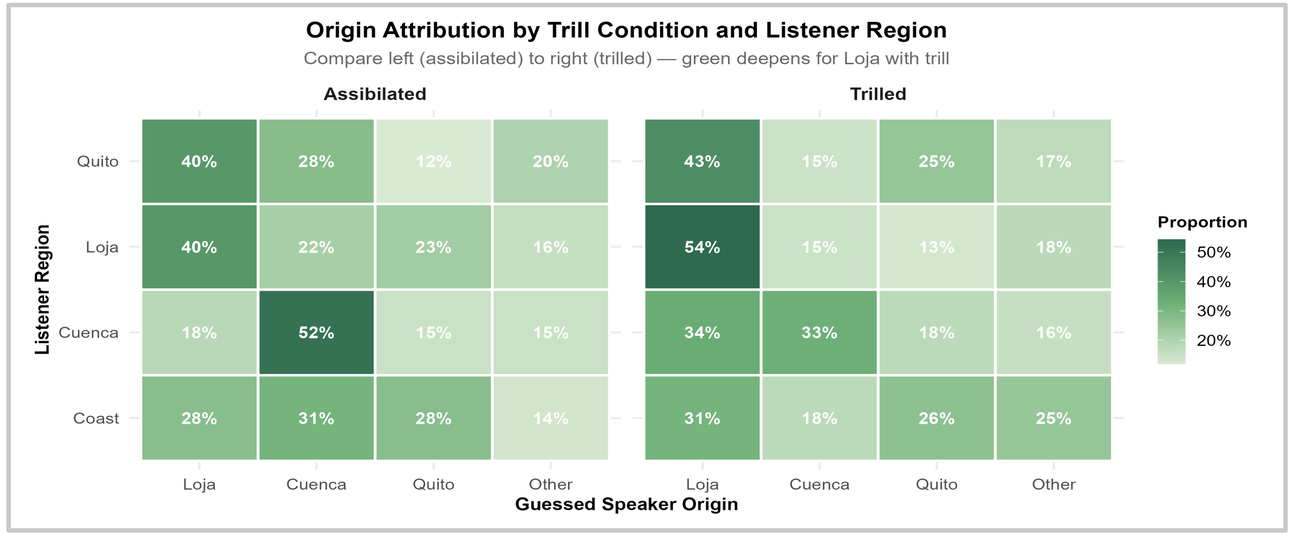
Reported Status

Model Preparation:

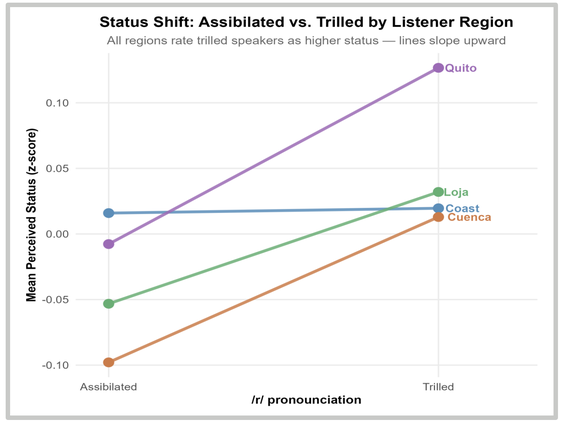
- Inverted femininity scale: masculinity = 7 - fem
- Created status composite row: Means of scaled class, urban, edu.
- Performed factor analysis on 6 listener rating numeric variables, the strongest was a status factor that combined education, class, and urbanism.



Origin Findings



Status Findings



Model 4: Does hearing a trill raise perceived status?
The trill effect was significant ($t = 2.61$)
Finding: Trilled speakers were rated as significantly higher status than assibilated speakers across all regions.

Conclusion

- Trill nearly doubled the odds of correctly identifying a speaker asLojano (Model 2, $p < 0.001$), functioning as a strong regional identitymarker specifically associated with Loja.
- Trill alone does not significantly predict broader origin attribution — listeners likely relied on a wider range of phonetic and linguistic features beyond trill alone when identifying speakers.
- Cuenca listeners showed the strongest trill-to-Loja association ($p = 0.038$), suggesting geographic proximity to Loja heightens sensitivity to trill as a regional cue.
- Lojano and non-Lojano listeners did not evaluate trilled speakers differently on status, confirming that trill prestige is perceived consistently regardless of listener origin.
- The perception of trill carried stronger social implications than geographic ones, suggesting /r/ variation in Ecuadorian Spanish functions primarily as a marker of social prestige.
- Overall, trill and assimilation had a much greater impact on perceived social status than on perceived origin.